

Handling imbalanced data

one class has far more rows than another

Change the data resampling

1. Random undersampling
drop majority rows

2. Random oversampling
copy minority rows

3. SMOTE
create new minority rows

Change the ensemble balanced samples

4. Balanced random forest
every tree sees a
balanced sample

Change the learning cost-sensitive learning

5a. Class weights
mistakes on the
rare class cost more

5b. Custom loss function
write the cost
of each mistake